

Snehal Prabhudesai

Ph.D. Candidate, Computer Science & Engineering, University of Michigan
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RESEARCH INTERESTS

I am an interdisciplinary scholar in Human-centered AI, HCI and Decision Science. My research shows that strong AI performance alone is insufficient for real-world impact, as people and organizations ultimately determine AI adoption and use. I employ technical, cognitive and organizational lenses to ensure AI is adopted responsibly and used effectively in high-stakes domains such as Healthcare, Education and Climate Change.

EDUCATION

Ph.D. in Computer Science & Engineering, University of Michigan, Ann Arbor 2026 (expected)
Dissertation: Computing and Communicating AI Uncertainty for Appropriate Reliance in High-stakes Human-AI Decision-making
Advisors: Nikola Banovic, Xun Huan, Arvind Rao
Committee: Q. Vera Liao, Jessica Hullman, Richard Gonzalez

M.S. in Electrical & Computer Engineering, University of Michigan, Ann Arbor 2020
GPA: 3.8/4.0

B.Tech in Electronics Engineering, K. J. Somaiya College of Engineering, Mumbai 2018
GPA: 8.75/10.0

AWARDS & HONORS

Rackham Graduate Student Research Grant (\$3000) 2025

CSE Honors Competition — HCC Runner-up 2024

Rackham Conference Travel Grants (\$2550) 2023, 2024

SIAM Travel Award (\$1000) 2023
invited to write an article for SIAM News

CSE Service Award for Excellence (\$1000) 2023
awarded to 5 of 279 PhD students

PUBLICATIONS

Conference Proceedings and Journal Articles

- [C.1] **“Here the GPT made a choice, and every choice can be biased”: How Students Critically Engage with LLMs through End-User Auditing Activity**
Snehal Prabhudesai, Ananya Kasi, Anmol Mansingh, Anindya Das Antar, Hua Shen, Nikola Banovic
ACM Conference on Human Factors in Computing Systems (CHI 2025). [pdf]
(Media coverage by CSE News, Michigan Daily and UNESCO’s Technology and AI in Education Unit)
- [C.2] **Understanding Uncertainty: How Lay Decision-makers Perceive and Interpret Uncertainty in Human-AI Decision Making**
Snehal Prabhudesai, Leyao Yang, Sumit Asthana, Xun Huan, Q. Vera Liao, Nikola Banovic
ACM Conference on Intelligent User Interfaces (IUI 2023). [pdf]
- [C.3] **Lowering the Computational Barrier: Partially Bayesian Neural Networks for Transparency in Medical Imaging AI**
Snehal Prabhudesai, Jeremiah Hauth, Dingkun Guo, Arvind Rao, Nikola Banovic, Xun Huan
Frontiers in Computer Science, Special Topic: Responsible AI in Healthcare, 2023. [pdf]
(Multiple invited talks at Society of Industrial and Applied Mathematics)

- [C.4] **Stratification by Tumor Grade Groups in a Holistic Evaluation of Machine Learning for Brain Tumor Segmentation**
Snehal Prabhudesai, Nicholas Chandler Wang, Vinayak Ahluwalia, Xun Huan, Jayapalli Rajiv Bapuraj, Nikola Banovic, Arvind Rao
Frontiers in Neuroscience, Special Topic: From Prototype to Clinical Workflow, 2021. [pdf]
- [C.5] **A Bayesian Approach for Quantifying Data Scarcity when Modeling Human Behavior via Inverse Reinforcement Learning**
 Tahera Hossain, Wanggang Shen, Anindya Antar, **Snehal Prabhudesai**, Sozo Inoue, Xun Huan, and Nikola Banovic
ACM Transactions on Computer-Human Interaction (TOCHI 2023). [pdf]
- [C.6] **Federated Learning enables Big Data for Rare Cancer Boundary Detection**
 Sarthak Pati, Ujjwal Baid, . . . , **Snehal Prabhudesai**, . . . , Spyridon Bakas, Bjoern Menze.
Nature Communications, 2022. [pdf]
 (International collaboration across 71 institutions and 6 continents, largest-ever federated learning study)
- [C.7] **Towards Fair Decentralized Benchmarking of Healthcare AI Algorithms with the Federated Tumor Segmentation (FeTS) challenge.**
 Maximilian Zenk, Ujjwal Baid, . . . , **Snehal Prabhudesai**, . . . , Klaus Maier-Hein, and Spyridon Bakas.
Nature Communications, 2025. [pdf]
 (International collaboration across 32 institutions for real-world performance evaluation of tumor segmentation algorithms)

In Submission

- [S.1] **Rethinking the Last Mile Gap: A Field Study on Challenges of AI Integration in Clinical Practice**
Snehal Prabhudesai, Avery Maddox, Santhoshi Krishnan, Errol Colak, Susanne Gaube, Nikola Banovic
to ACM Designing Interactive Systems (DIS 2026)
 (in collaboration with University of Toronto and University College London.)
- [S.2] **“Designing technology is not just a technical problem, but also a people problem”: Multidisciplinary Team Dynamics and their Effects on the Design of Disaster Risk Communication Tools**
Snehal Prabhudesai, Sarah Whittaker, Sabine Loos, Alicia Carmichael, Richard Gonzalez, John Ziker, Xun Huan, Nikola Banovic
to ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW 2026)
 (Multidisciplinary, multi-institutional study with 14+ climate scientists, psychologists, and flood emergency managers)
- [S.3] **Trained to feel ten feet tall and invincible: How institutional culture affects flood resilience at two U.S. military installations**
 Sarah Whittaker, **Snehal Prabhudesai**, John Ziker
to American Ethnologist
 (Top-5 journal for social and cultural anthropology)
- [S.4] **Reimagining How Flood Warnings Can Inform Decision-Making and Community Actions**
 Vinh Ngoc Tran, Xun Huan, Nikola Banovic, Jeff H. Bednar, Shannon Marie Bergt, Chen Cheng, Anindya Das Antar, Francina Dominguez, Kevin Gray, Richard Gonzalez, Brian Jewett, **Snehal Prabhudesai**, Deffi Putri, Sudhansu Rath, Emily L. Shroyer, Sarah H. Whitaker, John P. Ziker, Valeriy Y. Ivanov
to Nature Perspective
 (authors listed alphabetically)

Extended Abstracts (Refereed & Juried)

- [EA.1] **A Physics–Human Integrated Decision-making Framework for DoD Installations under Flood Risks**
 Xun Huan, Valeriy Yu Ivanov, Francina Dominguez, John Ziker, Nikola Banovic, Richard Gonzalez, Brian F. Jewett, Chen Cheng, Vinh Tran, **Snehal Prabhudesai**, Deffi Putri, Kevin Gray, Sudhansu Rath, Remi Masterson, Sarah Whitaker
American Geophysical Union Fall Meeting Abstracts, 2024
 (National Research Council community to advance earth and space science)

Technical Reports

- [TR.1] **A Taxonomy for Understanding and Identifying Uncertainty in AI-Generated Responses**
Snehal Prabhudesai, Daniel G. Goldstein, Jake Hofman, David M. Rothschild
SSRN Electronic Journal, 2024
 (presented internally at Microsoft NYC to 30+ attendees)

- [TR.2] **Reducing Uncertainty in Multijurisdictional Urban Stormwater Systems under Climate Extremes**
 John Ziker, Sarah H. Whitaker, Remi Masterson, Shannon Bergt, Xun Huan, Richard Gonzalez, **Snehal Prabhudesai**,
 Nikola Banovic, Valeriy Ivanov
SSRN Electronic Journal, 2025
 (in submission to Nature Cities)

Doctoral Consortium

- [DC.1] **Communicating Uncertainty for Appropriate Reliance on Flood Decision Support Systems**
Snehal Prabhudesai
ACM Conference on Computing and Sustainable Societies (COMPASS 2025).

RESEARCH EXPERIENCE

- Microsoft Research** 2023
Research Intern, Computational Social Science Group, New York
 Supervised by: David Rothschild, Jake Hofman, Daniel Goldstein
- Conducted simulated user experiments and analyzed 1024 LLM responses to conceptualize a taxonomy of uncertainty in LLMs.
 - Submitted a full research paper to Behavioural Science and Policy 2024: [\[arXiv preprint\]](#)
- Uncertainty Quantification and Scientific Machine Learning Group** 2019
Research Assistant, Mechanical Engineering, University of Michigan
 Supervised by: Xun Huan
- Conducted sensitivity analysis on a 7.8 million-dimensional U-Net model for brain tumor segmentation.
 - Submitted a poster to ICASSP 2020 and full research paper to MLHC 2021.
- Systems Imaging and Bioinformatics Lab** 2019
Research Assistant, Michigan Medicine, University of Michigan
 Supervised by: Arvind Rao
- Trained a CNN on 65,088 MRI dataset from the Medical Segmentation Decathlon Challenge.
 - Calibrated model with Platt scaling and computed AU-ROC, sensitivity and dice scores.

INVITED TALKS AND PANELS

- Evaluating Educational Use Policies of Large Language Models** 2021
 LEAD Now, Student Organization at UM School of Public Policy
- Computing and Communicating Uncertainty in Human-AI Decision-making** 2023
 SIAM Conference on Uncertainty Quantification
- Transparency via Uncertainty in Medical AI** 2023
 SIAM Conference on Computer Science and Engineering
- Partially Bayesian Neural Networks** 2022
 SIAM Conference on Uncertainty Quantification
- The Potential of Algorithmic Handoffs** 2022
 Human Machine Collaboration in a changing world
- Panel on Human-Machine Collaboration in Healthcare** 2022
 Algorithmic Futures Policy Lab Series
- Panel on Introduction to Graduate Studies** 2021
 EECS 598: Intro to CSE Graduate Studies

TEACHING EXPERIENCE

Teaching Assistant

[A.1] **EECS 493: User Interface Development**

Department of Computer Science

4-credit upper-level undergraduate course with class size of 220+

Held biweekly Office Hours, managed 7 aides and 4 graders. Taught HTML, CSS, JS, Python. Student rating: 4.8/5.0

[A.2] **BDSI22: Big Data Summer Institute**

Department of Biostatistics

Six-week interdisciplinary undergraduate training program with class size of 20+

Taught project-based Introduction to Machine Learning Course. Created course materials and Github repository.

Invited Guest Lectures

[L.1] **Classification Methods in Machine Learning**

for PUBHLTH 383: Data Driven Solutions In Public Health

4-credit upper-level undergraduate course with class size of 100+

Taught machine learning and statistics for public health decisions. Invited to teach for 3 consecutive years.

[L.2] **Holistic Evaluation for Responsible AI in Healthcare**

for SI 542: Introduction to Health Informatics

3-credit graduate course with class strength of 50+

Taught analysis, design, implementation, and evaluation of healthcare decision support tools

MENTORING

Gilian Soekawan	currently undergraduate, University of Washington	2025
Yara El-Tawil	currently PhD, Computer Science, University of Michigan	2025
Ananya Kasi	moved on to PhD, Computer Science, Georgia Tech	2024
Anmol Mansingh	moved on to Machine Learning Engineer, Instacart	2023
Rui Nie	moved on to PhD, Biostatistics, University of Michigan	2022
Leyao (Hannah) Yang	moved on to MS, Data Science, Columbia University	2021
Vinayak Ahluwalia	moved on to MS, Health Policy, University of Pennsylvania	2020
Dingkun Guo	moved on to MS, Robotics, Carnegie Mellon University	2019

SKILLS

UX Methods: Usability testing, sketching & prototyping, think-aloud, user interviews, Wizard-of-Oz

Quantitative: Field experiments, surveys, data analysis, statistical modeling

Computational: Machine learning, Bayesian statistics, uncertainty quantification

Languages/Tools: Python, R, JavaScript, TensorFlow, PyTorch, OpenCV, MATLAB

SERVICE

To Community

Associate Chair, CHI 2026. Understanding People: Mixed Methods Subcommittee 2025-2026

Session Chair, CHI 2025. Explainable AI 2025

Program Committee. Workshop on Bi-directional Human-AI alignment, ICLR 2025 2025

Accessibility Student Volunteer, CHI 2022 2022

Reviewer. Nature Scientific Reports, ACM FAccT, CHI, CSCW, UIST 2022-present

Received 8 Special Recognitions for Outstanding Reviews

To Department

Coordinator. Friday Night AI at Ann Arbor District Library <i>Organized 5+ panels with 20+ experts on social topics for public AI literacy initiatives</i>	2025
Coordinator. Michigan Science Centre with CSE AI <i>Designed pop-up exhibits and prepared quizzes on Generative AI capabilities for public AI literacy.</i>	2025
Board Member. ECSEL+ <i>Organized practice prelims in CS Department for 15+ students across 4 areas of CS.</i>	2021-2022
Member. CSE Outreach and Program Evaluation Committee <i>Represented CS Department at TAPIA Conference on Diversity in Computing with 1800+ attendees</i>	2021-2023
Founder. CSE@OUTDoors <i>Grassroots organization for queer representation across staff, faculty and students in Computer Science.</i>	2021-2023

MEDIA COVERAGE

- [M.1] **Researchers holding critical views instead of falling into AI companies' academies.**
Chief of UNESCO's Technology and AI in Education Unit [\[article\]](#) (May 2025)
- [M.2] **CSE celebrates grad student achievements at 2025 recognition reception**
University of Michigan Computing Research News [\[article\]](#) (April 2025)
- [M.3] **Medical Centers Tap AI, Federated Learning for Better Cancer Detection**
NVIDIA News [\[article\]](#) (September 2024)
- [M.4] **U-M GPT, one year later**
The Michigan Daily [\[article\]](#) (November 2024)
- [M.5] **Federated Learning—A Solution for Democratizing Data for Cancer Research?**
National Cancer Institute [\[article\]](#) (January 2023)
- [M.6] **Machine learning approach detects brain tumor boundaries**
National Institutes of Health [\[article\]](#) (December 2022)
- [M.7] **Intel and Penn Medicine Announce Results of Largest Medical Federated Learning Study**
Intel Press Releases [\[article\]](#) (December 2022)

PROFESSIONAL MEMBERSHIPS/AFFILIATIONS

Association for Computing Machinery (ACM)
ACM Special Interest Group on Computer–Human Interaction (SIGCHI)
Society of Industrial and Applied Mathematics (SIAM)

REFERENCES

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jhullman@northwestern.edu

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Senior Principal Researcher, Microsoft Research
davidmr@microsoft.com